Ethan Gonzalez

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EDUCATION

University of Pennsylvania, *School of Engineering and Applied Science* — Philadelphia, PA

- MSE in Robotics and BSE in Mechanical Engineering (Expected May 2028); Mathematics Minor — August 2024 – Present
- *Cumulative GPA*: 3.79 / 4.00
- *Activities and Societies:* Project Wavefront, UPenn ASME, UPenn Engineers Without Borders, St. Anthony Hall Fraternity

Bishop O'Dowd High School — Oakland, CA

- *Cumulative GPA:* 4.44 / 4.00 | *SAT Scores:* Math 800 | Writing 740 — August 2020 – June 2024
- *AP Scores:* Calculus BC 5 | Physics 5 | Statistics 5 | Chemistry 5 | Spanish Language 5 | US Government 5 | English Language 4

PROFESSIONAL EXPERIENCE

Modular Robotics Lab (GRASP Lab), *Robotics Research Assistant,* Philadelphia, PA — September 2026 – Present

- Developing mechanical systems and testing twisted-string actuators for ABOT, a self-assembling modular robotics platform

Pyka, *Mechanical Engineering Intern,* Alameda, CA — May 2026 – August 2026

- Designed and flight-tested 2 electric-motor cooling architectures, reducing operating temperature by ~20% and supporting increased continuous power capability
- Redesigned the aircraft spray-fluid fill system and developed a custom pressure-relief valve, raising allowable operating pressure from 15 to 25 psi through flow testing and rapid prototyping
- Designed and flight-tested 3 FPV camera assemblies, completing FOV calculations and wire-harness routing to enable BVLOS operations and reduce required operators by 1
- Redesigned wing and spray-boom hardpoints and hangers for corrosion resistance and manufacturability, and designed airfoil-shaped carbon composite spray booms using CFD (AirShaper), reducing aerodynamic drag power losses by 200+ W; completed 20+ GD&T drawings, tolerance stackups, and ECOs

Astra Space, *Manufacturing Engineering Intern,* Alameda, CA — May 2025 – August 2025

- Designed 15+ CAD (NX) components and test stands for Hall-effect satellite thrusters, presenting and defending design decisions in 5+ technical reviews and quality checks
- Developed and executed 300+ torque-optimization trials, enabling ~44% reduction in fastener weight via improved tightening
- Designed a custom anode-cathode assembly tool and a Swagelok krypton feed system, improving assembly precision and yield
- Applied Design for Manufacturability (DFM) principles and implemented GD&T standards (10+ drawings)

Project Wavefront, *Mechanical Engineer,* Philadelphia, PA — July 2025 – Present

- Led CAD (Creo) design of the 1st undergraduate-designed additively manufactured rotating detonation rocket engine system
- Optimized for L-PBF additive manufacturing, practicing top-down modeling and FOS engineering, and conducting failure analysis
- Conducted ANSYS Fluent CFD and structural FEA of hot-wall and cooling-channel sections, applying thermal gradients and coolant pressures to evaluate temperature-dependent yield FOS and guide design iterations

High School Honors Science, Math & Engineering Program, *Researcher,* Michigan State University, MI — June 2023 – December 2023

- Performed ~12,000 computational simulations evaluating directional electric-field effects on PFAS-water interactions, revealing field configurations that produced a 200% increase in modeled remediation efficiency
- Utilized supercomputer to calculate simulations, performed comprehensive analysis on molecule diffusion and bonding, and presented findings weekly to research team
- 3.5% admit rate; Research paper submitted for publishing and invited to present research March 2024

LEADERSHIP EXPERIENCE AND SERVICE

Engineers Without Borders, *Project Engineer for Penn Chapter,* Philadelphia, PA — January 2025 – Present

- Developed an accessibility ramp for Grace City Church in Philadelphia to support community events and address inaccessibility
- Design selected by engineering advisor for construction; collaborating on site measurements, CAD modeling, and design iteration
- Engineering a well system for an educational farm in The Gambia, including hydraulic, water demand, and budget calculations

Educate and Elevate Charity, *Founder,* Alameda, CA — August 2021 – July 2024

- Advocate, inform, and raise funds to address gender, socioeconomic, and racial inequities within access to STEM education
- Developed and provided 6-week computer science curriculum and instructed dozens of girls
- Raised $3,000+ for Chromebooks; Partnered with Girls Inc. and Scientific Adventures for Girls

Varsity Cross Country and Track, *Team Captain,* Oakland, CA — August 2020 – June 2024

- *Captain:* Attend weekly meetings with coaches, lead warmups and workouts, introduce new runners to the sport, and help organize practices and races for 40+ teammates 6 days/week
- Established and successfully met team objectives by creating a positive environment while holding members accountable

HONORS AND AWARDS

American Society of Mechanical Engineers, *Executive Board Treasurer of Penn Chapter,* Philadelphia, PA — September 2024 – Present

- Managed $2,000+ in funding and coordinated communication for 6 professional development events connecting mechanical engineering professionals with the UPenn engineering community
- Supported engineering outreach through quality programs advancing human well-being

Regeneron Science Talent Search, *Scholar,* Society for Science — January 2024

- Top 300 scholar in the Regeneron Science Talent Search, receiving $2,000 to further my PFAS research
- Nation's oldest and most prestigious science and mathematics competition for high school seniors

SKILLS AND INTERESTS

Software: CAD (SolidWorks - CSWA, Siemens NX, PTC Creo, AutoCAD) | CFD (ANSYS, AirShaper) | FEA (ANSYS, SolidWorks) | Microsoft Office
Other: GD&T | Python, C, & HTML | Microcontrollers (Arduino, ItsyBitsy) | Lathe & Mill (Mastercam) | Conversational Spanish | Jira, Confluence, Odoo, Teamcenter
Interests: Skiing | Climbing | Running | Mountain Biking | Traveling | Welding | 3D Printing | Laser Cutting